databricks

MERIDIAN HOTELS & RESORTS

Acting on Booking Cancellations Before Rooms Go Unsold

Revenue Management Cancellation Intelligence on Databricks

Meridian Hotels & Resorts is a fictional company. Booking data is the public Hotel Booking Demand dataset: real bookings from two hotels, 2015–2017.

THE HEADLINE

Cancellations cost an estimated $16.7M in room-night revenue across two hotels

$16.7M

Estimated lost room-night revenue

37.0%

Cancellation rate (44,224 canceled)

119,389

Bookings, City Hotel and Resort Hotel, 52 arrival months

Source: gold_hotel_month. Estimated lost revenue = ADR × canceled room-nights; an estimate, not RevPAR.

WHERE IT IS LOST

City Hotel accounts for about two-thirds of estimated lost revenue

City Hotel — 79,330 bookings

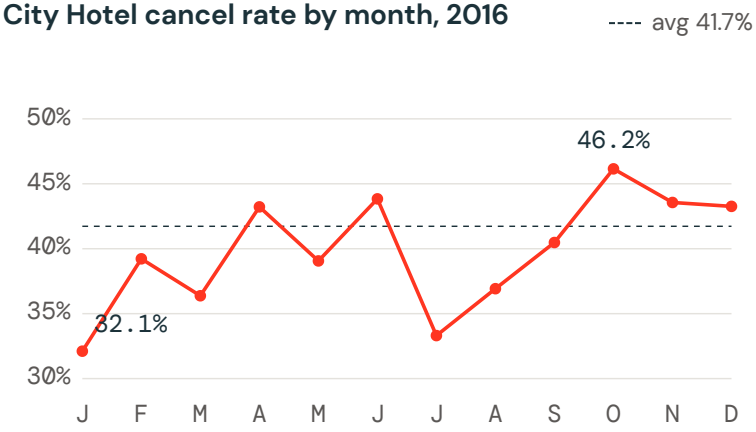
$10.9M est. lost revenue

41.7% cancel rate

Resort Hotel — 40,059 bookings

$5.8M est. lost revenue

27.8% cancel rate



Peak month overall: City Hotel, May 2017, ~$879K est. lost revenue.

Source: gold_hotel_month. Lost revenue figures are estimates (ADR × canceled room-nights).

THE DECISION PROBLEM

Which bookings will cancel, and what should we do about each one?

A revenue manager needs a ranked list of at-risk reservations and a clear lever for each.

01

Request a deposit

Ask high-risk guests for a deposit to confirm intent.

02

Controlled overbooking

Hold a backup booking against rooms likely to be released.

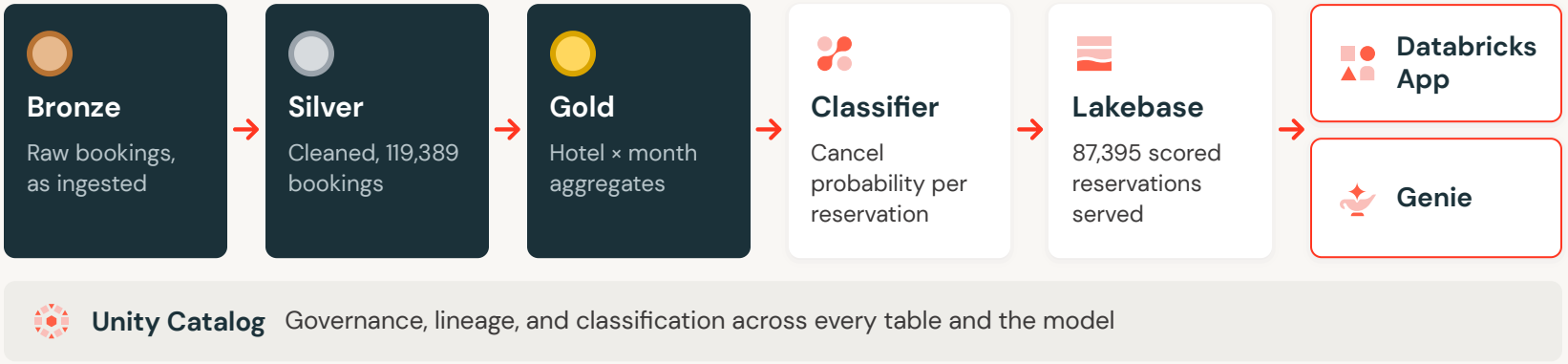
03

Reconfirmation nudge

Contact the guest to reconfirm, or release the room early.

HOW IT WORKS

One governed pipeline from raw bookings to scores in the revenue manager's hands



Source: hotel_booking_dev catalog. Lakebase serves a deduplicated snapshot (87,395 of 119,389 rows).

HOW GOOD THE MODEL IS

About three in four flagged bookings do cancel, tested on arrivals the model never saw

ROC-AUC	PR-AUC	F1
0.84	0.81	0.67

TEMPORAL SPLIT

The model learned from earlier arrivals and was tested on the latest ~20%, the same way it will be used on future bookings.

Gradient-boosted classifier, hotel_cancel_classifier v1

Precision 0.76



Of bookings flagged as likely to cancel, about 76 in 100 do.

Recall 0.60



Of bookings that do cancel, the model flags about 60 in 100.

Higher threshold
Fewer flags, higher precision, lower recall

Lower threshold
More cancellations caught, more false alarms

Source: hotel_cancel_classifier v1 @champion, held-out temporal test set.

RISK TRIAGE

26,814 high-risk reservations (~22%) are the actionable set



High · p ≥ 0.70

26,814

Average cancel probability 0.92

Medium · 0.40–0.70

14,281

Average cancel probability 0.54

Low · p < 0.40

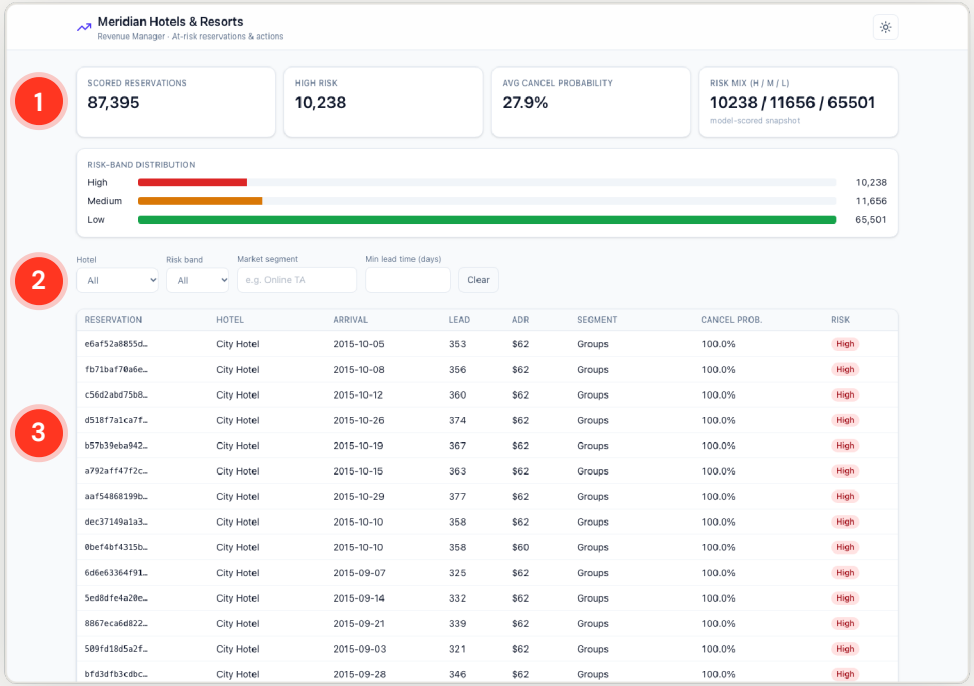
78,294

Average cancel probability 0.15

Source: reservation_risk, all 119,389 scored reservations.

THE REVENUE MANAGER'S VIEW

The at-risk list, ranked and filterable



1 KPI header

87,395 scored, 10,238 high risk, 27.9% average cancel probability.

2 Risk filters

Hotel, risk band, market segment, and minimum lead time.

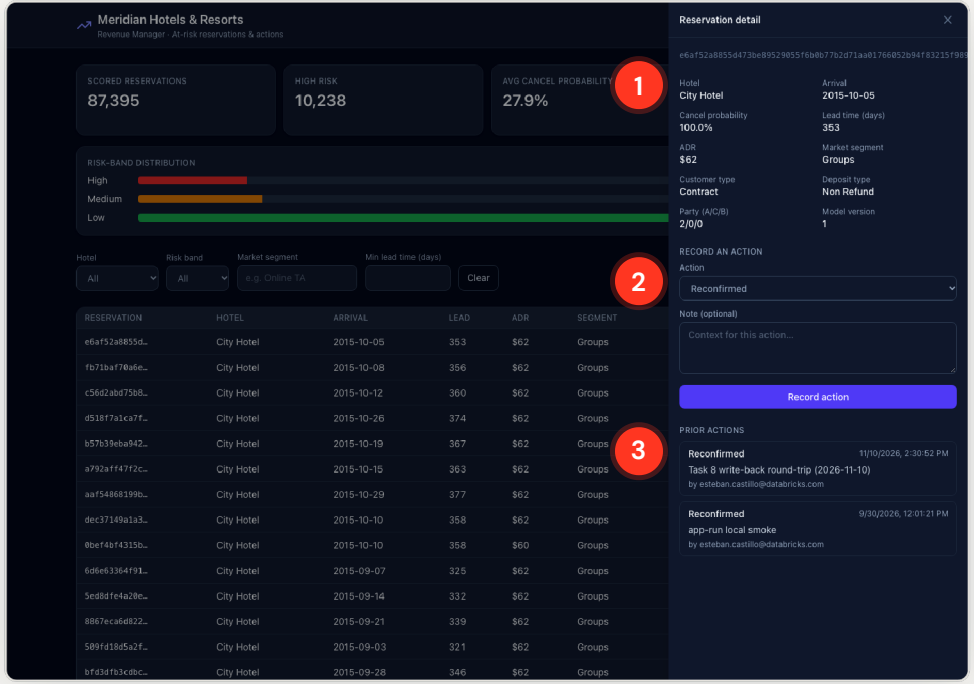
3 Sorted at-risk list

Highest cancel probability first, with arrival, lead time, ADR, and segment.

Source: live app, Lakebase snapshot (deduplicated; full table holds 119,389 / 26,814).

TAKING ACTION

Every decision is recorded against the reservation



1 Reservation detail

Cancel probability, lead time, ADR, deposit type, and model version.

2 Recorded action

Reconfirm, deposit requested, overbooking backup, release, or note.

3 Audit trail

Prior actions with timestamp and the user who took them.

Source: live app, reservation detail drawer with Lakebase write-back.

GOVERNED BY DESIGN

Controls are built into the data, not added by the app

PII masking

Column masks on country, agent, and company, with classification tags.

Per-property access

Row-level security so each hotel sees only its own reservations.

Lineage

Bronze, silver, gold, and model scores traced in Unity Catalog.

Audit trail

Every recorded action is stored with the user's identity.

EXPOSURE AND NEXT STEPS

$16.7M is the exposure we can quantify today. Recovery is what the pilot measures.

QUANTIFIED NOW (ESTIMATED, FROM DATA)

$16.7M

est. lost room-night revenue: City Hotel $10.9M, Resort Hotel $5.8M

26,814

high-risk reservations (~22.5% of 119,389 bookings), the actionable set

Method: est. lost revenue = ADR × `stay nights`, for canceled bookings. A room-night proxy, not RevPAR.

ILLUSTRATIVE SENSITIVITY

ILLUSTRATIVE

Arithmetic on the $16.7M exposure, not a forecast or result.

1% recovered	5% recovered	10% recovered
~$167K	~$836K	~$1.67M

PILOT WILL MEASURE

01 Recovered rooms and avoided lost revenue at one property

02 Overbooking accuracy: rooms recovered without walk-outs

NOT MEASURABLE WITH THIS DATASET

RevPAR and occupancy impact

Both need room inventory data, which this dataset does not include.

Next: pilot at one property, measure recovery, then add room inventory for RevPAR and occupancy.

APPENDIX

Methodology and sources

Data source	Public Hotel Booking Demand dataset (Antonio, Almeida & Nunes): real bookings from two hotels, 2015–2017. The Meridian company framing is fictional.
Estimated lost revenue	`adr × stay_nights` when canceled, else 0. A room-night proxy and an estimate, not RevPAR.
Model and split	HistGradientBoostingClassifier (sklearn Pipeline), hotel_cancel_classifier v1. Temporal split; latest ~20% of arrivals held out for test.
Evidence	Committed under `docs/evidence/` in the project repo; live queries against `hotel_booking_dev` (2026-11-10).